

Europe has a solid basis for battery and electric vehicle manufacturing growth

Europe is successfully building up battery and EV manufacturing capacity, though mainly thanks to foreign investment

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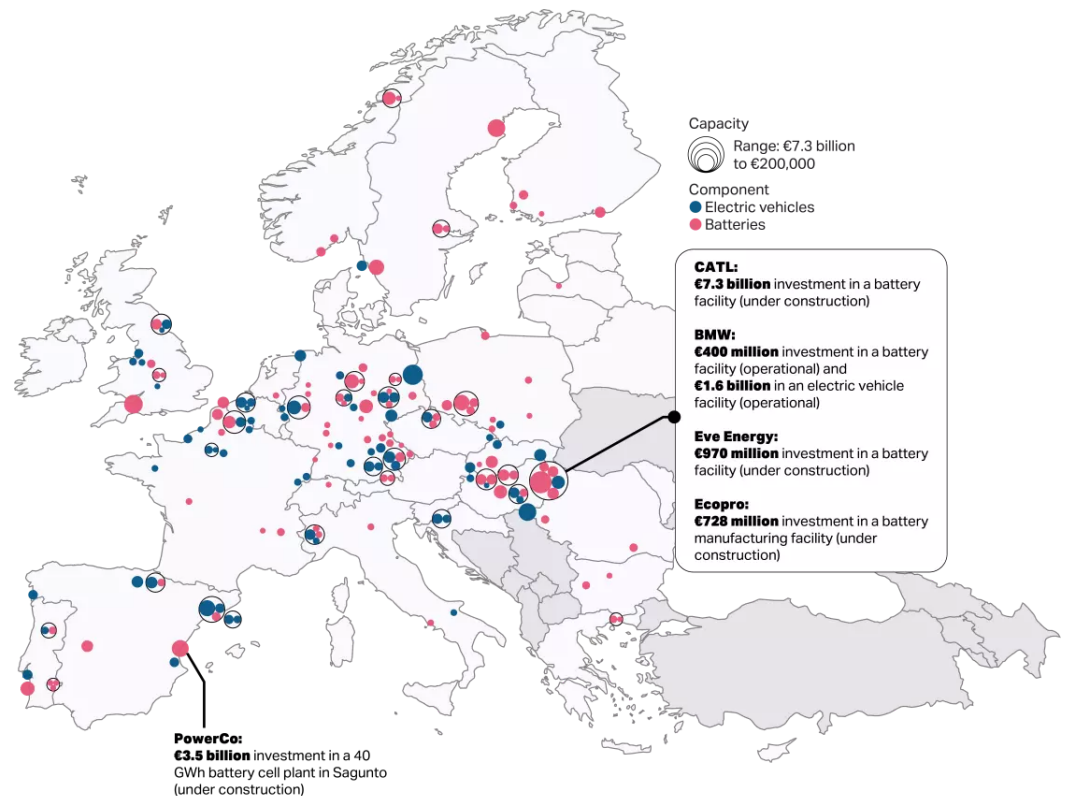
Batteries and electric vehicles (EVs) contribute to Europe's energy security, climate policy and industrial strategy. Stationary batteries help to stabilise electricity grids, while EV adoption helps reduce reliance on imported oil and reduces the one-sixth of global carbon dioxide emissions from road transport¹. The two technologies accounted for two-thirds of the \$700 billion global clean technology market in 2023, showing their relative importance compared to other cleantech (IEA, 2024). In the United States, for which comprehensive data is available, investment in battery and electric vehicle manufacturing outstrip other clean-tech investment².

The European Union is using policy to shape domestic battery and EV markets. Rules limiting the average fleet-wide carbon emissions from new cars and vans (Regulation (EU) 2019/631) have increased the incentives for manufacturers to supply and market electric vehicles, boosting uptake. Significant fiscal support at national and European levels has been used to attract battery and EV manufacturing investment. Meanwhile, the EU has imposed tariffs on Chinese-produced EVs, while European EVs now face higher US tariffs.

Battery and EV investments therefore show how an industrial transformation is unfolding alongside the clean energy transition. Figure 1 shows these investments since 2017, with bubble sizes proportionate to the value of investments. Since January

2017, we estimate that companies have invested €38 billion in European battery manufacturing facilities and a further €34 billion in electric vehicle facilities (Bruegel, 2025)³.

Figure 1: Battery and electric vehicle manufacturing investments, 2017-2025



Source: Bruegel (2025). Note: covers 292 investments in 113 battery and EV manufacturing facilities; investments include new facilities and the expansion or conversion of existing sites; investments are tracked from when they are first publicly announced, construction begins and when projects are completed or, occasionally, cancelled; see Bruegel (2025).

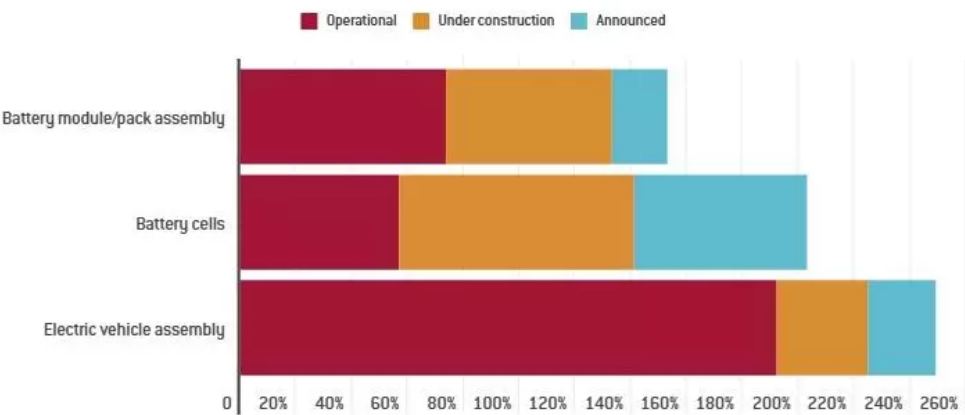
This analysis addresses three questions. First, what have investments delivered in terms of battery and EV manufacturing capacity and how does that compare with domestic demand? Second, who owns these facilities and how important has foreign investment been? Third, how are investments distributed across Europe and how has this changed over the past eight years?

What has €72 billion in investment delivered?

Each investment is tied to an annual manufacturing capacity: numbers of EVs that can be assembled, or, for batteries, cell fabrication and battery pack assembly stages. As

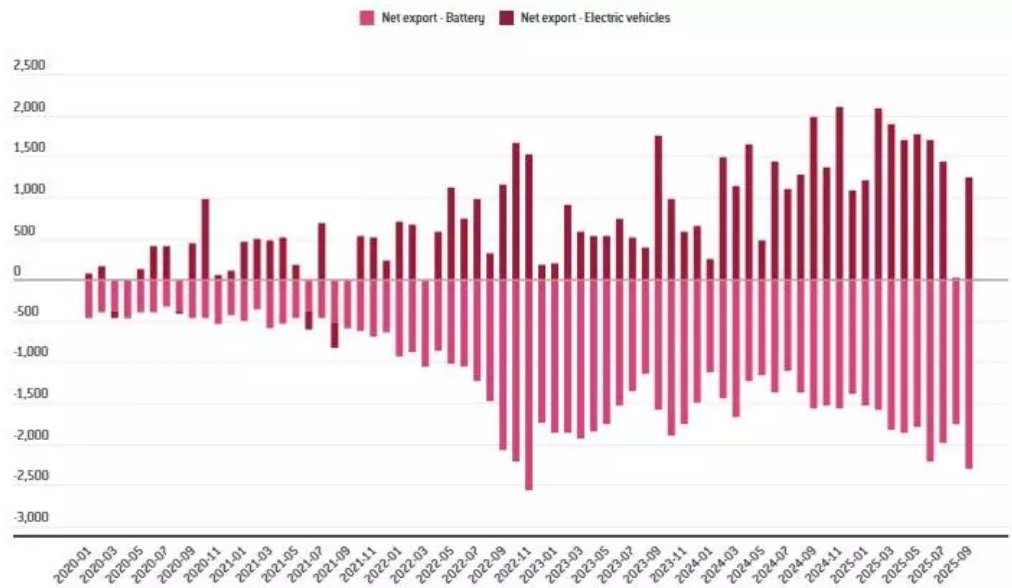
of September 2025, we estimate that European factories had the capacity to produce 4.6 million electric vehicles (including pure battery and plug-in hybrid) and 251 gigawatt hours (GWh) of battery cells⁴ annually. This compares to 2024 annual demand of 2.3 million electric vehicles and 410 GWh of batteries, of which roughly 90 percent are for EVs and the remainder are stationary batteries. Factories in Europe thus have the capacity to produce twice as many EVs as domestic demand, and to meet approximately two-thirds of domestic battery cell demand (Figure 2).

Figure 2: European manufacturing capacity expressed proportionally to 2024 demand



Source: Bruegel (2025).

A proper evaluation of investments should cover not only delivered capacities but also the utilisation of this capacity. Such analysis is complicated because production data is closely guarded by industry. Trade data offers an approximation of production. On a net basis, the EU imports batteries and exports electric vehicles (Figure 3). In 2024, net exports of EVs from the EU were worth €14 billion, while net imports of batteries amounted to €17 billion.

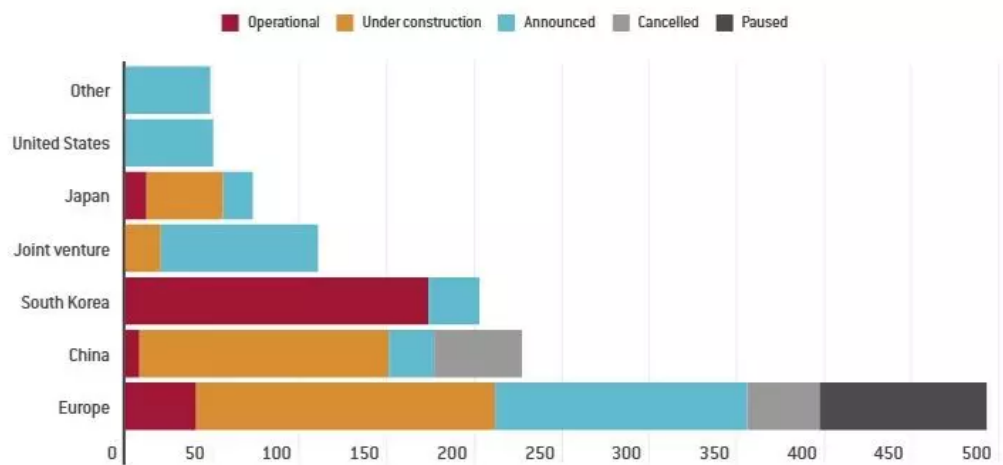
Figure 3: EU monthly net trade in electric vehicles and batteries (€ millions)

Source: Bruegel (2025).

The main destinations for EU exports of EVs over the past five years have been the United Kingdom (one-third of all exports) and the US (one-fifth). Exports to Türkiye have been rising and reached 10 percent of EU EV exports in 2025, while Norway and Switzerland receive around 9 percent and 6 percent, respectively. The EU's position as a net exporter of EVs is threatened by the August 2025 EU-US joint agreement which set a 15 percent tariff on US vehicle imports⁵. EV exports to the US dropped substantially, by almost 40 percent, in July-September 2025 compared to the same period in 2024.

Who owns Europe's manufacturing facilities?

Foreign direct investment, particularly from South Korea, has been crucial for building EU battery manufacturing capacity. Three companies (LG Energy, SK Innovation and Samsung SDI) were early innovators in lithium-ion technologies and today own four-fifths of operational battery cell manufacturing capacity in Europe (Figure 4). LG Energy, with European Investment Bank support, built Europe's largest battery cell manufacturing facility in Wrocław, Poland, with 86 GWh or 35 percent of the continent's capacity. SK Innovation began construction of the €1.6 billion Ivánca (Hungary) facility in March 2022, and Samsung SDI has invested over €1.5 billion into the Göd (Hungary) facility since 2016.

Figure 4: European battery cell capacity (GWh) by region of owner's HQ and status

Source: Bruegel (2025).

While South Korean investment has slowed recently, substantial Chinese investment is arriving. Most notably, in May 2025, CATL, a Chinese company and the largest car battery producer worldwide, began construction in Debrecen (Hungary) of what may become Europe's largest battery cell facility with a potential capacity of 100 GWh/year and investment reaching €7.3 billion.

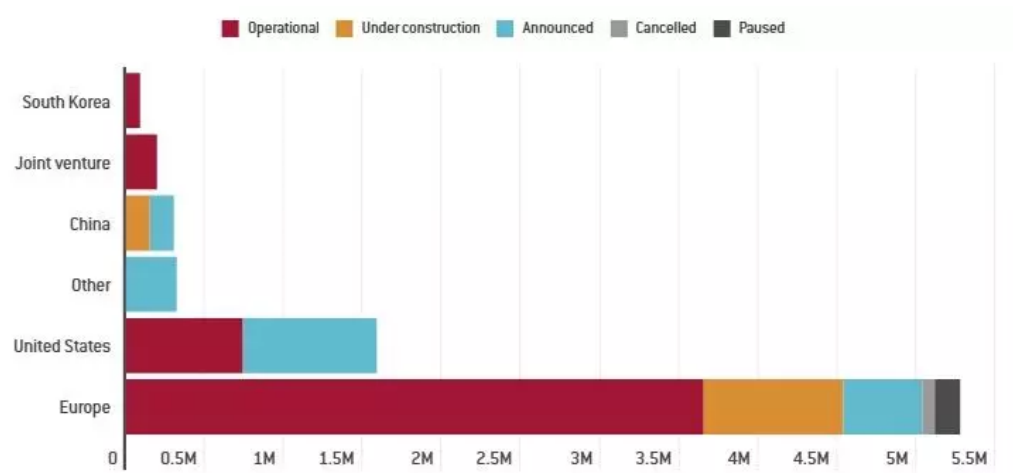
The EU has tried to develop home-grown battery champions. Northvolt's €4 billion investment in Skellefteå (Sweden), launched in late 2018, secured a \$5 billion loan backed by the European and Nordic Investment Banks to expand the site – still the largest green loan ever raised in Europe⁶. While Northvolt subsequently declared bankruptcy (Tagliapietra and Trasi, 2024), the 16 GWh Skellefteå plant began operations and is being sold to US firm Lyten⁷. France is home to the two other European-owned operational battery cell facilities. Verkor operates an 8 GWh facility in Dunkirk, while the Automotive Cells Company joint venture runs a 13 GWh plant in Douvrin. Further projects are under construction by PowerCo (a Volkswagen subsidiary) and Volvo, with the latter at time of writing searching for a new partner to complete its Gothenburg (Sweden) facility, following Northvolt's collapse.

For electric vehicles, foreign companies are less dominant. Established European automakers, including Volkswagen, BMW, Renault and Stellantis, own most EV capacity, partly because many investments involve retrofitting or expanding existing factories (Figure 5). In 2020, Volkswagen completed the €1.2 billion conversion of its

Zwickau (Germany) facility to produce only electric models and began a €1 billion electric retrofit of its Emden (Germany) plant.

The largest single EV investment in Europe is the €5.8 billion by Tesla in their facility in Grünheide (Germany), where construction began in January 2020. The facility has the capacity to assemble 375,000 EVs annually. Chinese company BYD is building a €4 billion facility in Debrecen (Hungary), with an expected annual capacity of 150,000 once complete.

Figure 5: European electric vehicle manufacturing capacity by region of owner’s HQ and status

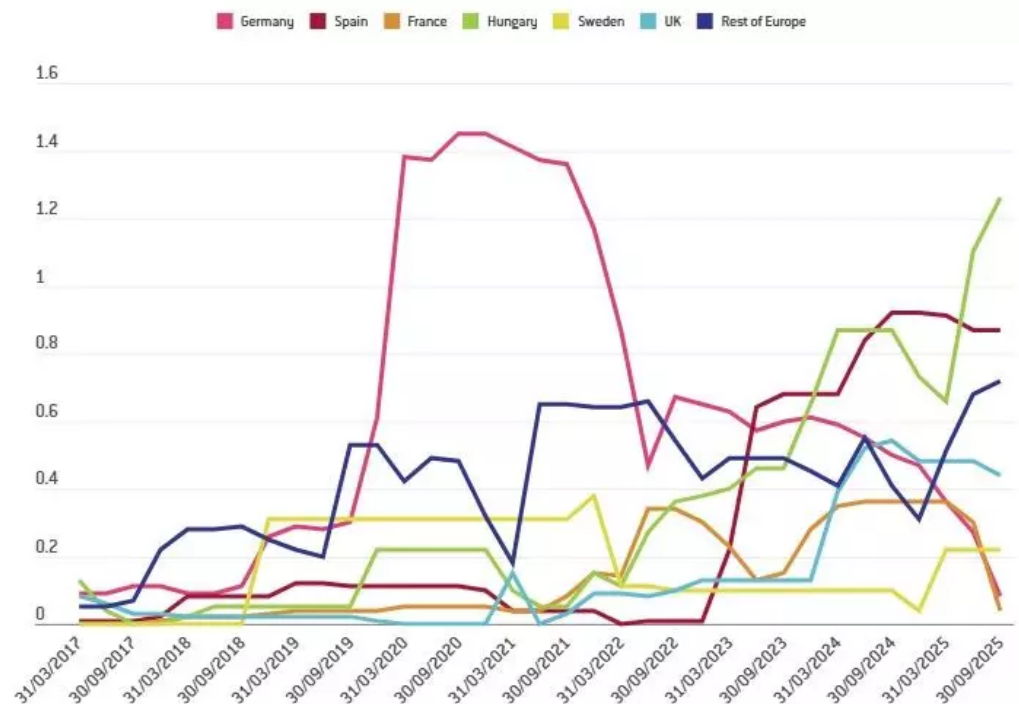


Source: Bruegel (2025).

Is Europe’s industrial map shifting?

Overall, since 2017, Germany has attracted the largest volume and number of investments in EVs and batteries in Europe. However, in 2025, Hungary and Spain are Europe’s leading investment destinations (Figure 6).

Figure 6: Ongoing investments into battery and EV manufacturing in Europe, € billion



Source: Bruegel (2025). Note: announced investment volumes are distributed uniformly between the dates on which an investment begins and finishes according to construction timelines; for example, a €1.2 billion investment beginning in January 2021 and finishing in December 2021 would count for €100 million in investment in each month of 2021.

Investment in Hungary comes predominantly from Chinese companies. This includes the CATL €7.3 billion battery cell and BYD €4 billion EV investments mentioned above, but also Eve Energy (€950 million, battery cell) and EcoPro (cathodes, €700 million).

Whilst CATL has announced a joint venture with Stellantis to invest over €4 billion in a battery plant in Zaragoza, ongoing investments in Spain are not driven by Chinese companies but rather the Volkswagen Group. In 2023, Volkswagen subsidiary PowerCo began construction of a potentially 40 GWh battery cell plant in Sagunto, near Valencia. Volkswagen itself is investing €1 billion to prepare its Navarre facility for EV production. Meanwhile Seat, another Volkswagen subsidiary, is investing €3 billion in its Pamplona plant to prepare for further EV production.

Poland and Hungary are currently neck-and-neck leading in battery cell production capacity⁸. Together they account for over two-thirds of European capacity. While LG is

expanding its Polish facility, Chinese investment in Hungary means the country is on course to become Europe's leading battery manufacturer. Germany is far ahead of any other European country in terms of EV capacity, with Spain and France ranking next⁹.

Success so far – and keeping it that way

The EU now boasts a solid industrial base for the manufacturing of both batteries and EVs. This base can already meet a large part of domestic battery demand, and the EU has become a net exporter of EVs.

EU vehicle CO₂ limits have been crucial for establishing stable demand expectations. Any proposal to weaken the EU's 2035 deadline for CO₂ limits to reach zero¹⁰ should be rejected by European governments and the European Parliament. This weakening would unfairly penalise companies that have already invested billions of euros into EV and battery production and would discourage future investment. It would impede the core objective of the Clean Industrial Deal – the European Commission's plan to marry decarbonisation and economic competitiveness¹¹ – and harm the long-term competitiveness of the EU car industry. It would also seriously undermine Europe's reputation as a global climate leader. The rapid cancellation of projects in the US following the dismantling of policies that supported battery and EV investments throughout 2025 offers a warning¹².

Foreign investment has been responsible for most battery investment, initially from Korean companies but today increasingly from China. The construction of these plants has created European jobs and longer value chains by enabling investment by European companies in downstream EV facilities. Continued openness to foreign investment is important for battery supply chains and for other clean technologies.

While Europe has healthy battery and EV manufacturing capacities today, demand for both will increase and additional supply will be needed. Since 2022, there has been a shift in manufacturing investment away from Germany and towards Hungary and Spain. The energy transition involves a restructuring of manufacturing supply chains, and it is inevitable that some industrial relocation will be economically efficient. Policy should embrace rather than fight this.

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Endnotes

- 1 See Hannah Ritchie, 'Cars, planes, trains: where do CO₂ emissions from transport come from?' Our World in Data, 6 October 2020, <https://ourworldindata.org/co2-emissions-from-transport>.
- 2 See Rhodium Group, 'The US Clean Investment Monitor', undated, <https://rhg.com/energy-climate/data-and-tools/clean-investment-monitor/>.
- 3 'Europe' includes the United Kingdom and Norway. In this analysis, by battery and electric vehicle investment, we mean investment in manufacturing of batteries and EVs.
- 4 GWh units express the energy storage capacity of batteries produced. One GWh of battery production is approximately sufficient to produce 15,000 EVs.
- 5 See statement of the European Commission DG for Trade and Economic Security, 'Joint Statement on a United States-European Union framework on an agreement on reciprocal, fair and balanced trade', 21 August 2025, https://policy.trade.ec.europa.eu/news/joint-statement-united-states-european-union-framework-agreement-reciprocal-fair-and-balanced-trade-2025-08-21_en.
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- 8 See Bruegel (2025): <https://european-clean-tech-tracker.bruegel.org/technology/batteries/manufacturing>.
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- 10 Jordyn Dahl and Gerardo Fortuna, 'Commission set to severely weaken 2035 combustion engine ban', Politico, 15 December 2025, <https://www.politico.eu/article/european-commission-2035-co2-emissions-climate/>.

- 11 Ben McWilliams and Simone Tagliapietra, 'Europe's Clean Industrial Deal: four priorities to fulfil its promise', First Glance, 29 September 2025, Bruegel, <https://www.bruegel.org/first-glance/europes-clean-industrial-deal-four-priorities-fulfil-its-promise>.
- 12 See Ben German, 'Clean-tech cancellations outstripped new investments in Q2', Axios, 28 August 2025, <https://www.axios.com/2025/08/28/us-tech-projects-growth>

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